Flood damage assessment in urban areas. Application to the Raval district of Barcelona using synthetic depth damage curves

Marc Velasco

CETaqua Water Technology Centre, Carretera d'Esplugues 75, 08940, Cornellà, Barcelona, Spain (Email: mvelasco@cetaqua.com)

Àngels Cabello

CETaqua Water Technology Centre, Carretera d'Esplugues 75, 08940, Cornellà, Barcelona, Spain

Beniamino Russo

Aqualogy Urban Drainage Direction, Diagonal Avenue, 2011, 08018, Barcelona, Spain.

Technical College of La Almunia (EUPLA, University of Zaragoza), Mayor Street, 50100, La Almunia de Doña Godina, Zaragoza, Spain.

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This paper presents a detailed flood damage assessment in the Raval district of Barcelona. The implementation of a new 1D-2D coupled model is used to obtain flood depths, and new stage damage curves are developed to estimate the direct tangible damages. The curves have been validated using data from surveys and actual reported damages to the Spanish re-assurance. Finally, combining hazard and vulnerability levels by using a GIS-based toolbox, the expected annual damage of the area is obtained. This enables the determination of the critical points of the district in terms of flooding impacts, and highlights the need to implement strategies to cope with these impacts.

Keywords: Urban flooding; impact assessment; depth damage curves.

1. Introduction

Urban areas are, due to the concentration of population and economic activities, one of the most sensitive regions to natural hazards. Trends show that world's population is moving to cities: currently 50 % of the global population lives in urban areas, and by 2030 at least 61 % of the world's population will be living in cities (IBRD/WB 2009). In Europe, such values present even more extreme situations: 83% of the population is expected to live in cities by 2050 (EC 2010).

The concentration of people in cities increases their opportunities as well as their vulnerabilities to natural hazards and climate change impacts (Djordjević *et al*. 2011). Consequently, the assessment of urban flood impacts is an issue of high interest, specially taking into account the framework that the European Flood Directive (EC 2007) has developed. The Directive defines flood risk as "the combination of the probability of a flood event and of the potential adverse impacts on human health, the environment, cultural heritage and economic activity associated with a flood event". Other definitions of the same concept available in the literature agree that "flood risk is the probability of a

certain flood event, in combination with its impact" (DEFRA/EA 2006). Therefore, flood impacts are a crucial part of the flood risk equation.

In the last few years and in agreement with the European Flood Directive, the traditional "flood control approach" has been partially replaced by "flood risk management" (Merz *et al*. 2010a). The focus in this paradigm shift is put in flood risk instead of flood hazard. Therefore, nowadays vulnerability is considered as important as hazard. Flood damage assessments are gaining more importance within this evolving context of decision-making for flood risk management (Merz *et al*. 2010b).

Although the first research initiatives started working on this field almost forty years ago (Penning-Rowsell and Chatterton 1977), there is still a lot of work to do related to this issue. All approaches of flood damage analysis contain uncertainties due to the inaccuracies and generalisations used (Messner *et al*. 2007). The lack of data aggravates this problem, being estimation methods very rough.

Nevertheless, a lot of flood damage models have been created recently, aiming to ease the calculation of impacts in both qualitative and quantitative ways (Jongman *et al.* 2012). Seven flood damage models were compared in two different case studies. Results conclude that some more improvements are required in this field, with the final goal to produce a European framework that applies best practices from existing models while taken into account the regional characteristics of each area.

The work presented in this paper aims to establish a framework to assess flood damages in urban areas from the very begging. The final output obtained will be flood damage maps, allowing to identify the most critical areas of the studied area. By calculating the costs for different return periods, the expected annual damage (EAD) of the studied domain can be obtained, being able to express to which extent is the area

affected by floods.

To demonstrate this methodology, the whole process is applied in an urban area affected by floods, where no previous damage assessments have been carried out: the Raval district of Barcelona.

2. Methodology

2.1. Context

Flood impacts may be very diverse depending on the type of damage and the method used to assess them. Therefore, it is important to state the assumptions made, the categories considered and the approaches used to assess these impacts.

The general classification used for flood damage makes a first distinction between tangible and intangible ones. Tangible damages are those which can be assessed in monetary terms (Smith and Ward 1998; Messner *et al.* 2007). The second common distinction is between direct and indirect. Typically, a direct damage is defined as any loss that is caused by the immediate physical contact of flood water with humans, property and the environment. Jonkman *et al*. (2008) referred to direct losses as those occurring within the flooded area, and indirect damages, outside the flooded area. In Table 1, a summary of the several categories with a few examples can be seen.

Regarding the scale of the study, Messner *et al*. (2007) proposed the following classification: macro, meso, and micro scales. In the latter, studies apply an object-oriented approach, and consequently damages are calculated for single properties. Hence, as opposed to other scales, data requirements are high, the models used must be very detailed and land-use information should be collected at a building scale.

Flood damage assessment is an issue that has not been deeply studied in Spain. There are some novel experiences (Francés *et al*. 2008), but in general, flood risk assessments, such as in the Catalan flood risk management plan, INUNCAT (GENCAT

2006), tend to assess the impacts using methodologies from other countries. For the INUNCAT case, the methodology developed in the USA (USACE 2000) was directly applied to Catalonia, without adapting it. That is why experiences from other countries (Penning-Rowsell *et al*. 2010, Nascimento *et al*. 2007, Kok *et al*. 2004, Reese *et al*. 2003, Merz *et al*. 2010b, Messner *et al*. 2007) were also reviewed in order to undertake this study.

Estimation of damages caused by flooding normally focus on flood depth. That is why depth damage or stage damage curves have been adopted in multiple locations around the world as the most commonly used technique to assess flood impacts.

Depending on the information available and the goals of the assessment, there are several types of depth damage curves (Merz *et al*. 2010). In some cases, when there is a lot of data available, empirical curves can be created by interpolating a function with the depths and costs relationships (Nascimento *et al.* 2007). Alternatively, synthetic curves use the expertise in flood impact assessment to create the curves via a what-if analysis (Penning-Rowsell *et al*. 2010). Additionally, depth damage curves can also be classified as absolute or relative, depending if the cost is given in economic terms or as a percentage of the total value of the affected asset.

Thieken *et al.* (2005) introduced that there are two types of damage influencing factors: the impact parameters and the resistance parameters. The first ones reflect the specific characteristics of a flood event (such as water depth, flow velocity, etc.) and the second ones represent the properties of the affected assets (such as building type or materials, emergency measures used, etc.). In Merz *et al.* (2010) an extensive review of all the damage parameters was carried out. It was concluded that, with the exception of flood depth, most of these damage influencing factors are neglected in damage modelling.

Sometimes, flow velocity can be a very important factor because it can increase the probability of structural building damage due to lateral pressure, scouring, etc. (Merz *et al.* 2010). On the other hand, flow velocity, due to the related hydrodynamic force, is an important point to be considered for pedestrian (Russo et al. 2013a) and vehicular stability (Shand *et al*. 2010).

However, this impact parameter is strongly influenced by several resistance parameters. In a study undertaken after hurricane Katrina (Pistrika and Jonkman 2009), it was identified that variations in building type implied important changes in the resistance of buildings. In this study it was identified that most of the residential properties that were completely destroyed in New Orleans were made of wood, and proved to have limited resistance to flood impacts.

2.2. Proposed methodology

As stated previously, the goal of this work is to develop a methodology to obtain flood damages in urban areas from the very begging, being able to produce flood maps and calculate the EAD of the studied area.

Since this study pretends to assess impacts from a starting point in which there is no data at all, only one typology of damages will be considered. Given that in urban areas industrial uses are not so important, and considering that the main communication roads are generally surrounding urban areas, only direct damages will be considered. In addition, since the goal is to calculate the damages in economic terms so the EAD can be obtained, only the tangible ones are included in this assessment.

The spatial variability of urban damages is very high. The impacts induced by urban floods can differ a lot from one street to the next one. Therefore, in terms of the spatial scale, assessments in urban areas clearly require a micro-scale study.

Flood damage assessment is recent research field that has not been deeply studied in some countries such as Spain. Considering this as a starting point, and assuming that there is a lack of historical damage data, synthetic absolute depth damage curves must be developed. These curves will not take into account flood velocities, because the construction typology and materials of European (and more particularly Spanish) buildings are less affected by this variable.

Summarizing, this impact assessment will consider direct tangible damages at a micro-scale level, by using synthetic depth damage curves. In order to carry out a damage assessment, and particularly for this one, three key elements are required: depth damage curves, detailed flood depth maps and land-use maps.

Furthermore, in order to ease the calculation of the final flood damage maps, a GIS-based toolbox has been developed (Hammond *et al.* 2012). This toolbox enables to automatize the three following steps, increasing the speed of the post processing of data and so, easing the simulation of several events:

(1) Read the value of water depth in each building from flood hazard maps;
(2) Interpolate this value in the stage damage curve to obtain the relative cost;
(3) Multiply the relative cost by the area, obtaining the total damage value per each block.

Finally, a shape file with the total cost values is obtained, being able to calculate the total costs that have been caused by the extreme rainfall event. This process can be easily repeated for several flood-driven events.

Using this toolbox and data described in the following sections, synthetic rain events for several return periods can be simulated. These results will be used in order to calculate the EAD of the studied area, which is used to better understand the potential damages caused by a hazardous event.

EAD is an estimate of the average flood damages computed over a number of years (Arnell 1989) and it is obtained by integrating the relationship between the expected costs for an event and its probability (Dawson *et al*. 2008).

Since it is difficult to accurately define the relationship between the probability of a flood event and the costs it would cause, the most common methodology to calculate EAD consists on simulating several events of different return periods (which means that they have a different probability associated) and calculate the damages for each case. Then, by calculating the area under the curve that these points define, the EAD can be obtained.

3. Case study description

Barcelona, with a population of 1,620,943 within its administrative limits on a land area of 101.4 km^2, is located in Catalonia, on the Northeast coast of the Iberian Peninsula, facing the Mediterranean Sea, on a plateau limited by the mountain range of Collserola, the Llobregat river to the south-west and the Besòs river to the north east (Figure 1). The average population density of the city is 15,985 inhab./Km^2, which approximately becomes 19,200 inhab./Km^2 if the forest area of Collserola mountain is not considered.

Barcelona presents a classic Mediterranean climate with cool winters and even warm summers, occasionally suffering heavy rainfalls of great intensities and flash floods events. The yearly average rainfall is 600 mm, but the maximum intensity in 5 minutes corresponding to a return period of 10 years is 202.47 mm/h and it is not rare that 50 % of the annual precipitation occurs during two or three rainfall events. The Intensity Duration Frequency (IDF) curves were recently updated on the basis of a rainfall series data of 81 years (from 1927 to 1992 and from 1995 to 2009). On the basis of these IDFs new project storms with several return periods were obtained for the design of the sewer network (Rodríguez *et al*. 2013).

The morphology of Barcelona presents areas close to Collserola mountain with high gradients (with an average of 4% and maximum values of 15-20%) and other flat areas near to the Mediterranean Sea with mild slopes (close to 0-1 %) or even hollow areas susceptible to floods (Figure 2).

The Raval district of Barcelona is located in a hollow area of the city (Figure 1 and Figure 2). With almost 50,000 inhabitants in an area of 1.09 km^2, it is one of the most densely populated areas in Europe (approx. 44,000 inh./km^2). The district area is highly impervious with several vulnerable elements (such as schools, hospitals, museums, historic buildings, etc.).

This area suffers from flooding problems when heavy storm events occur. These problems are caused by the excess of surface runoff and the poor capacity of the sewer system in some of the upstream basins in the city. In addition, the hydrological response time of Raval district catchment is very short (less than 30 minutes). As a result, there is significant hazard for the vehicular and pedestrian circulation, as well as economic damages in terms of goods and properties.

4. Data

In order to implement the methodology described in section 2 in the Raval case study, several datasets have been compiled and treated in order to obtain flood damage maps and finally calculate the EAD of the area.

As stated previously, to calculate the EAD several events of different return periods must be simulated. In this study, five events have been selected in order to calculate this variable. These events correspond to a really extreme event (T = 100 years), a precipitation event that starts to create some damage (T = 1 year) and three intermediate

ones. Since the Barcelona sewer network is predominantly designed to cope with a 10-year return period event, this will be the third return period used (T=10). The other two return periods are some more intermediate values of the three ones described, which are 5 and 20 years. Such rain events have been simulated by using the hyetographs developed from the IDF curves (Rodríguez *et al*. 2013).

In the following sections, the information and procedures used to create the depth damage curves, the land-use maps and the flood maps are presented.

4.1. Depth damage curves

Since there were no curves specifically representing the flood damages in the studied area, synthetic absolute depth damage curves were developed in the frame of this study. Such curves are used to obtain costs for a certain water depth relative to the extent flooded. Then, multiplying the obtained value by the affected area of the building, the damage cost at a block scale is acquired.

These curves have been created for different types of land-use. Taking into account the main uses identified in the case study area, six different categories have been defined (warehouses and parking areas; commercial; residential; hotels and leisure; public and cultural buildings; and sites of interest - such as museums, churches or historical buildings). Then, via a what-if analysis and using flood expertise acquired from past flood events, a stage damage curve for each type of building and economic activity has been obtained. These final stage damage functions are composed by two independent curves.

The first ones, related to the re-conditioning of the building (cleaning, painting the walls, changing the floors or doors, etc.), have been obtained through the expertise gained

in past local floods. This study was developed with the collaboration of an independent flood damage appraiser with expertise in the Barcelona city area (Lucena 2012).

The process to develop these curves is described following: (i) a standard building has been defined for each of the six land-use categories considered; (ii) the costs of the several cleaning and refurbishment works required have been assigned to different water depths; (iii) these costs have been normalized using the surface of the standard building defined; and (iv) adding the several categories the final building depth damage curve is obtained.

For example, the residential standard building is of 80 m^2, with three bedrooms, two toilets, one kitchen, a living room and an entrance hall. The cleaning and refurbishment works required are cleaning works, water pumps, brickwork, painting, wood works and refurbishment of the building systems. As it can be seen in Table 2, the absolute and relative damages for each category and depth are presented for a residential standard building.

The second component of the curves is related to damages to the contents that can be found inside the buildings. In order to create them, the same standard buildings defined for the building curves have been used. For each land-use category, several assets and goods have placed inside the standard buildings. Then, using the FloReTo tool (Manojlovic and Pasche 2010), a database of flood depth damage curves, and introducing there the Spanish market prices of the exposed assets, the contents curves have been obtained. Again, by normalising the values using the surfaces of the standard buildings, the relative depth damage curves have been created.

Using the same example as before, the main goods and assets considered for the residential building are furniture, clothes, electronic devices and domestic appliances. The damages for different depths to each contents category can be seen in Table 3. The

several curves developed as explained previously, for both buildings and contents, can be seen in Figure 3.

The relationship between building and contents damages strongly depends on the type of land-use considered. Whereas in households the building costs tend to be higher than the contents' (Thieken *et al*. 2005), this trend is not so clear in other land-uses. In the case of commercial use, flood losses are highly variable due to the differences depending on the kind of business considered (Gissing and Blong 2004). In this case, the methodology used considering the several standard buildings has led to curves that define, for the commercial land-use, the damages of contents up to five times greater than the building ones.

As it will be described in the next section, basements are also included in the land-use data sets. Therefore, such as the properties on the ground floor, the basement blocks will have a surface assigned to each land-use class. Consequently, it makes sense to relate them to depth damage curves that do not depend on the ground floor uses.

However, in general, damages in the basements are calculated using a single curve that represents both the ground floor and the basement, in which negative depths are used to obtain the damages in the basement (Luino *et al.* 2009, Scawthorn *et al.* 2006, McBean *et al.* 1988). In these approaches, the water fills the basement first, and then it reaches the ground level. In addition, the usage of negative depths is combined with the use of depreciated values for the basements. As stated by the US Federal Emergency Management Agency (FEMA 2009), the basement value could be expressed as percentage of the total value of the building.

Given that in this case the data availability and knowledge of the study area is higher, the methodology that has been followed is different. Instead of reducing the damages of the basements, a reduced flood depth has been implemented in the original

curves. Taking advantage of the flood event that occurred on 31st July 2011, a series of interviews were carried out in the affected area. Although the number of affected people was small, the answers provided very interesting information which was very useful for the development of the curves and the validation process. In order to establish the relationship between flood depths in the ground floor and the basements, information gathered during these interviews was used.

As it will be presented in section 4.2, basements on the case study area only represent a small part of the total surface. From the affected and interviewed properties, only 16% had a basement. From these, a 35% presented water on the basement but no water on the ground floor, a 20% had water on the ground floor, and the rest, a 45%, had water in both. Therefore, it seems that the relationship of the flood depths on one floor and the other strongly depends on the configuration of the property. Obviously, if the staircase going to the basement is next to the entrance, all the water will be flowing down there, but in other cases (such as having a step in front of the staircase) it may occur otherwise.

Given the uncertainties that this implies, and taking into account that this flood model is always considering flood water in the ground floor, the following methodology has been implemented here in order to try to represent the reality in a more accurate way: the water depth in the first basement will 0.5 times the water depth in the ground floor; and the subsequent basements will only have a water level of 10% the one in the ground floor.

The validation of the curves is crucial in order to assess their ability to represent the damages produced. As much information as possible regarding flood impacts in the studied area was gathered. Firstly, the information of the interviews previously mentioned was used for the validation process.

On the other hand, actual damage data from the Consorcio de Compensación de Seguros (CCS), the re-assurance that covers the catastrophic and extreme situations in Spain, were obtained for the same flood event in the Raval district.

The information from the two sources was used to adjust and validate the curves. The general trends and behaviours extrapolated from the surveys were taken into account; and the CCS damage data was compared to the simulated values.

From the surveys, two general conclusions were obtained: damages to contents are more important than damages to the building itself – consequently, the differences between the several pairs of curves from Figure 3 are justified; and flood depths smaller than a certain threshold do not cause any damages – and hence, the curves are zero until such value is reached.

When comparing the actual reported damages with the simulated ones using several thresholds, a few differences were found. With a 10 cm threshold, the affected area was larger than the one actually reported. That is why the threshold was increased to 15 cm, meaning that water depths smaller than this value would cause no damages.

Using the curves with this correction (Figure 3), the maps from Figure 4 were obtained. Here, the simulated damages (left) and the actual reported ones to the CCS (right), show very similar spatial distributions.

In terms of aggregated values in the whole district, the model is capable to obtain figures that are within the same order of magnitude. The simulated damages for the July 2011 event were of 750,002.9 € and the reported ones were 340,472.9 €. As observed, the simulated damages are more than two times bigger than the reported ones. Although this might seem inaccurate, it can be explained by the following reasons:

- Some of the flooded buildings may have not reported their damages to the CCS because they were low, the property was not insured or they were not aware that

they could be compensated. Many of these situations were observed with the interviews done for the July 2011 flood. Similar situations are observed elsewhere: in a study undertaken in the US (Cummins *et al*. 2010), it was found that for several natural hazards, the ratio between insured and total losses was in average smaller than 50 %.

- The simulated damages are assuming that no flood risk reduction strategies are being used. However, since some of the local population have suffered from previous flood events, some measures are taken to protect their assets, by placing wood gates in their doors, or moving goods to higher areas. The use of such strategies was identified from the results of the surveys done and in some other field visits. Some examples of this can be seen in Figure 5.
- The water depth inside the building has been assumed as the average of the maximum water depths of all the mesh elements in contact to each block. This hypothesis is conservative as it is an upper bound of the water that will actually be inside the building. Consequently, it is comprehensible that the costs obtained are also an upper bound of the actual damages, and thus the values achieved can be explained.

Besides the validation that has been presented, the developed depth damage curves have been compared to some other existing ones. In order to ease the comparison, the maximum damage values will be used (i.e. the value that represents the maximum costs that floods can imply, which for the developed curves corresponds to a depth value of 2 m or more).

Table 4 presents the values of several maximum damage values for the urban and commercial land-use categories. Results from the following models and studies are presented: the HAZUS Multi-Hazard software (FEMA 2009) from the United States; the

Multi-Coloured Manual (Penning-Rowsell *et al.* 2010) from UK; the work done by Francés *et al.* (2008) in Spain; the study for the IKSR, the International Commision for the Protection of the Rhine (LTV, 2003); the EC Joint Research Centre's (JRC) database of flood damage functions (Huizinga, 2007), particularized for Spain; and the Danish Coastal Authority results (Kystdirektoratet, 2004) for the Walden Sea case study.

The differences presented in Table 4 stress the importance of regional variations in asset values. This was also identified in Jongman et al. (2012), and it is one of the main reasons to establish a common European framework with regional flexibility to represent these changes. Still, the results presented in the table show that the curves developed in this study are in accordance with the several experiences created around the world.

As a conclusion, it can be said that the curves presented here have proven to represent accurately the flood damage processes in the Raval district. However, the intention is that in order to keep them valid, future work should consist on constantly validating or updating them (if needed) whenever more data is acquired.

Therefore, when future events imply damages in the studied area, the same process will be repeated to compare the simulated and reported values. On the other hand, if long series of records are obtained, instead of validating the curves for a given event, the costs could be compared in terms of mean values. Therefore, the mean annual damage recorded could be compared to the simulated EAD.

4.2. Land-use information

As stated previously, good quality and precision of data is crucial when carrying out a micro-scale study. Consequently, regarding land-use, a GIS map has been developed using data from the local land registry at block level (Figure 6, left). As the flood typology in the case study area consists in flash floods producing not so high water

depths (less than 2 m for storm with return periods of 100 years), only land-uses of the ground floor and basements have been added to the dataset.

For each block, more than one land-use type is possible, so the area related to the several land-use types is given. Multiplying these values by the relative damages obtained from the stage damage curves, the total costs of the block can be obtained. Additionally, it possible to calculate the risk map for goods and properties, which shows the maximum potential impacts in monetary units (Figure 6, right).

As it was previously mentioned, for this assessment, only six land-use types have been considered. Since the Raval district is a densely populated urban area, there are almost no green or industrial areas and no agricultural areas at all. Therefore, the uses that have been considered are: (1) warehouses and parking areas; (2) commercial; (3) residential; (4) hotels and leisure; (5) public and cultural buildings; and (6) sites of interest. This last category has been introduced because of the special importance of some of the buildings in this specific area, such as museums, churches or historical buildings with great cultural and historical value.

Since the water depth is going to be different depending on the floor level, the land-use datasets present this variable into two separate layers. It can be observed that whereas in the ground level the land-use types are evenly distributed, in the basements there is a predominance of warehouses and parking areas.

In addition, it is worth noting that basements only represent a small part of the total area. The added surface of the ground floors of the district reaches the 0.56 km^2 (which is a 76% of the total affected built area). On the other hand, the first basements occupy an area of 0.13 km^2 (i.e. 18%), and the subsequent basements represent a 6%, with 0.04 km^2.

4.3. Flood depths

The estimation of flood depth in a very accurate way is crucial for a micro scale assessment as this one. Therefore, there was a need for a coupled 1D/2D approach in order to take into account surface flows coming from upstream catchments and the interactions between the two drainage layers (known, respectively, as 'major system' formed by streets, sidewalks, squares, etc. and 'minor system' formed by the sewer network), as shown in Figure 7.

A detailed 1D/2D coupled model, simulating surface and sewer flows was developed using Infoworks ICM version 3.0 by Innovyze (2013). ICM solves the complete 2D Saint Venant equations in a finite volume semi-implicit scheme (Godunov, 1959) with a Riemann solver (Alcrudo and Mulet-Martí, 2005).

Special attention was paid to the hydraulic characterization of the inlet systems (representing the interface between surface and underground flows) using experimental expressions achieved in the Technical University of Catalonia.

In order to consider surface and sewer flows coming into the Raval District from upstream catchments, an extended area was considered in the study. Only main sewers were considered for these catchments, while main and secondary networks were taken into account for Raval District. The final model considered a total area of 44 km^2 with 3874 nodes, 241 km of total pipe length and 6 major storage facilities with a total capacity of 170,000 m^3.

A 2D mesh covered the whole analyzed domain with 403,822 triangles. Parks and other green areas were represented in the same 2D mesh, trough "2D infiltration zones" characterized by their specific hydrological, physical and geometric parameters, while buildings were represented as void areas. Runoff produced in the building areas was

estimated considering an approximation of single non-linear reservoir (whose routing coefficient depends on surface roughness, surface area, ground slope and catchment width) and directly conveyed into the sewer network. This goes in accordance with local practice in Barcelona, where roofs and terraces (approximately corresponding to 50% of the whole analyzed domain) are directly connected to the underground sewers.

Sewer model was calibrated and validated using data regarding 4 critical rainfall events occurred in 2011 and provided by CLABSA (the company that was managing the sewer network in Barcelona from 1992 to 2013) Control Centre. These data concerned 11 rain gages, 29 limnimeters and several time series related to real time control devices. Moreover, other data collected in the post events emergency reports (elaborated by policemen and firemen), and amateur videos recorded during the selected storm events were used to calibrate surface flow. Detailed information about the features of the model and, above all, the interactions between 1D and 2D layers is available in Russo *et al*. (2012 and 2013b).

The model only represents the water depth in the streets and green urban areas (as the buildings are considered as void areas for the surface layer and the runoff generated is directly conveyed into the underground sewers). As a consequence, the flood hazard maps obtained must be post processed in order to transform the water depth in the streets into the water depth inside the buildings.

Given that it is complicated to accurately determine the relationship between water depths inside the building and on the streets, a hypothesis has been used: the water depth inside the building is the result of calculating the average of the maximum water depths of all the mesh elements in contact to each block. This hypothesis is obviously an upper bound of the water depth inside the buildings because the buildings were considered void spaces in the model. Then, by assuming that the water depth inside the

buildings is the same as the level the streets, a surplus volume of water is considered. Consequently, this methodology "generates" a fictional amount of water inside the buildings. Therefore, the water level in the building will never be as high as the one on the street.

This overestimation of water depths obviously leads to an upped bound of the actual damages. Nevertheless, it has been preferred to work with an upper bound of the depths and damages, so the results are conservative and it can be sure that the simulation results are never below the actual value.

5. Results and discussion

Implementing the methodology and data presented in section 2 in the Raval district, results presented next are obtained. These results are given in two separate sections: a first one presenting the flood depths inside the buildings and a second one showing damages for the several simulated flood events.

5.1. Flood hazard

Using the design hyetographs developed from the IDF curves, five rain events of return periods of 1, 5, 10, 20 and 100 years were introduced as input to the 1D-2D coupled model. In Figure 8, depths generated by three of the simulated events can be seen. As mentioned earlier, the model outputs (i.e. water depth in the streets) have been converted into water depth inside the buildings in order to ease the calculation of the damages. Consequently, Figure 8 shows water only inside the buildings and not in the streets. In addition, the reader should note that water depths of less than 15 cm will cause no damage, as it has been stated in the section explaining the depth damage curves validation. Therefore, although the light blue colour is sometimes widespread, only the values over 15 cm will be considered for the damage model.

For an event of 1 year return period, flood problems are minor and since most of them are low depths, they do not cause a lot of damages. A similar situation occurs for a 5 year return period event, where the depths are still low, but the affected areas increase, especially affecting the western part of the district. For an event of 10 years of return period, these problems are more generalised, not only in the western part but also in the southern one. When increasing the severity of the rain event, this area enlarges and the water depths also increase. For a 20 year return period the both the depths and the flooded area are greater, and finally, for a 100 year return period event, the model presents a generalised flood in the district. This situation can be explained considering that the drainage system of Barcelona is generally designed for a return period of 10 years and hence, rain events of higher intensities produce a general failure of the system, being the flood problems widespread in the studied area.

Of course, the situation presented by the flood maps in Figure 8 is an upper bound of the actual situation. This is because the assumption of having the same depth inside the buildings than in the streets leads to an overestimation of the depths.

5.2. Flood damages

Using the flood maps presented in Figure 8 and the methodology described in section 2, flood damage maps are obtained for the studied area (Figure 9). As it can be seen in the right map of this figure, some blocks present extremely high values (more than 200,000 €).

From Figure 9, the same conclusion extracted from the hazard maps is reinforced: the southern and south-western parts of the district are the most vulnerable to floods. Due to the historic characteristic of this region with many buildings corresponding to the category “sites of interest”, the damages in this area are very high.

Finally, the EAD of the whole area has been calculated. As it has been stressed before, this value is of high interest because it helps to understand the potential impacts that can occur in an area due to a given hazard, i.e. floods in this case.

In Figure 10, the damage – probability curve for the Raval District is presented (for the detailed values, see Table 5). In it, the aggregated costs of the entire district are plotted against its probability. The area below this curve is the EAD, which has a final value of 1,116,078.5 €. As mentioned, this figure is an upper bound of the annual damage that may be caused by floods in the Raval District, which provides very interesting information about the magnitude of the impacts that can occur in the studied area.

When observing carefully Figure 10, it can be seen that there is clear shift in the log-linear relationship that exists between flood damages and the probabilities of the driving events. The breaking point that can be identified corresponds to a return period of 10 years, which is the design return period of the sewer network in Barcelona. This means that damage processes are associated with the way the system works.

For return periods smaller than 10 years, the sewer network is able to convey most of the water and hence, the log-linear relationship has a small slope. This means that the damages do not increase a lot with the intensity of the rainfall. On the other hand, the slope observed for return periods of more than 10 years is higher, implying that the damages will increase faster when the rainfall events are more severe. This can be explained by the fact that, for such return periods, the network is not functioning properly and so the flood damage processes are completely different than the other ones.

Some recent studies have identified similar patterns. Olsen *et al.* (2014) compared the results of calculating the EAD with three different methods. During this study, it was observed that a log-linear relationship existed in the damage – probability curves. However, they also found a shift in the curves, where smaller events followed one log-

linear relationship and the larger events followed another one. This breaking point was also identified as the design standard of the Danish sewer systems, which also corresponds to a 10 year return period (IDA 2005). The results showed that the shift in the log-linear relationship is more relevant than which method is used to perform the calculations.

In general, the minimum number of simulations considered for the calculation of the EAD is three (Penning-Rowsell et al. 2010, Messner et al. 2007). However, it is quite common that more simulations are used, like four (Aerts *et al.* 2013), five (Oliveri *et al.* 2000) or more (Zhou *et al.* 2012, Feyen *et al.* 2011, Elmer *et al.* 2012).

From what has been observed in this study, by choosing the exact three events, an accurate estimate of the EAD can be obtained. Therefore, using a really extreme event, a precipitation event that starts to create some damage and another one corresponding to the design return period of the sewer network, a rather precise figure of EAD can be obtained. This corresponds to the recommendations defined in the guidelines that were developed within the CORFU Project (Hammond *et al.* 2011).

6. Conclusions and further work

In order to improve the capacity to represent urban floods in the Raval District, a 1D-2D coupled model has been developed. The interface between the two drainage layers has been characterized through empirical expressions related to hydraulic performance of surface drainage systems. Calibration and validation of the model, as well as many other technical details can be found in Russo *et al*. (2012 and 2013b).

With the development of synthetic depth damage curves regionalized for the case study, an exhaustive economic impact assessment can be carried out when heavy storm events occur. Implementing the described methodology with a GIS-based toolbox, the EAD of the Raval district can be calculated. This enables the determination of the critical

points of the district in terms of flooding impacts and highlights the need to implement strategies to cope with these impacts.

From the flood damage maps obtained, the southern and south-western parts of the district are identified as the ones that are most vulnerable to floods, because of the presence of many sites of interest. The EAD of the whole area has a final value of 1,116,078.5 €, which means that in average, the magnitude of the flood impacts in the district is considerably high.

It is worth noting that damages calculated are an upper bound of the actual costs of the district, providing in this way a conservative value of the flood impacts that may occur.

The presented data and methodology establishes a very useful framework to assess flood impacts in urban areas. In this paper only the current impacts are assessed, but the framework is ready to be applied to assess the potential future damages, taking into account socio-economic and climatic changes, as well as implementing adaptation measures.

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Tables

Table 1. Categories of flood damage (adapted from Messner *et al*. 2007 and Jonkman *et al*. 2008).

		Measurement	
		Tangible	**Intangible**
Form of damage	**Direct**	- Physical damage to assets: • Buildings • Contents • Infrastructure • Agricultural land - Evacuation and rescue operations - Clean-up costs	- Fatalities and injuries - Diseases - Historical and cultural losses - Loss of ecological and environmental goods - Inconvenience
	Indirect	- Loss of industrial production - Traffic disruption - Emergency costs - Temporary housing of evacuees	- Societal disruption - Increased vulnerability of survivors - Undermined trust in public authorities - Psychological trauma

Table 2. Absolute and relative damages for a residential property of 80 m^2 and depths, classified by the different cleaning and refurbishment works required (Lucena 2012).

	Depths (m)									
	0.10		**0.20**		**0.30**		**0.50**		**≥1.00**	
	€	€/m^2	€	€/m^2	€	€/m^2	€	€/m^2	€	€/m^2
Cleaning works	120	1.50	150	1.88	260	3.25	632	7.90	632	7.90
Water pumps	0	0	0	0	0	0	0	0	450	5.63
Brickwork	0	0	0	0	80	1.00	320	8.13	650	8.13
Painting	0	0	0	0	0	0	950	11.88	950	11.88
Wood works	0	0	0	0	180	2.25	1200	15.00	1200	15.00
Building systems	0	0	0	0	0	0	200	2.50	450	5.63

Table 3. Absolute and relative damages for a residential property of 80 m^2 and depths, classified by the different content categories considered.

	Depths (m)									
	0.15		**0.20**		**0.50**		**1.00**		**≥2.00**	
	€	€/m^2	€	€/m^2	€	€/m^2	€	€/m^2	€	€/m^2
Clothes	0	0	153.3	1.92	383.3	4.79	1141.6	14.27	1750.0	21.88
Domestic appliances	0	0	310.0	3.88	930.7	11.63	1815.2	22.69	2050.0	25.63
Electronic devices	0	0	169.7	2.12	424.3	5.30	1005.0	12.56	1625.0	20.31
Furniture	0	0	386.1	4.83	1149.9	14.37	3785.3	47.32	5112.5	63.91

Table 4. Comparison of maximum damage values extracted from depth-damage curves around the world.

	Maximum damage values (€ / m^2)	
	Residential / urban land-use	**Commercial / industrial land-use**
Barcelona case study	185.87	260.58
HAZUS (US)	823.00	1156.00
MCM (UK)	714.00	1290.00
Francès *et al.* (Spain)	305.00	110.00
IKSR (Germany)	185.00	279.00
JRC (Spain)	400.00	225.00
Danish Coastal Authority (Denmark)	259.00	-

Table 5. Probabilities and damages for the five synthetic rain events simulated.

Return period (years)	**1**	**5**	**10**	**20**	**100**
Probability	1	0.2	0.1	0.05	0.01
Damage (€)	78,846	606,905	1,615,738	6,825,081	19,156,196

9. Figures

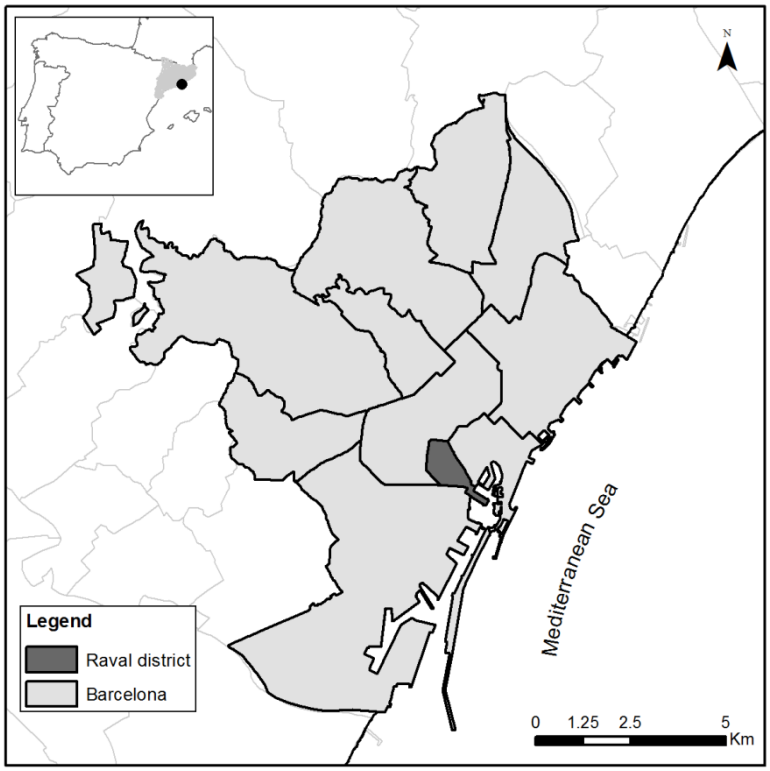


Figure 1. Administrative limits of Barcelona and representation of the case study area, the Raval district.

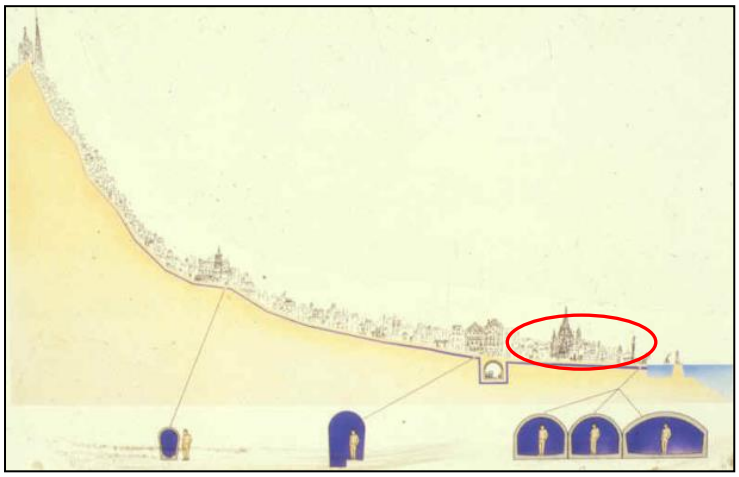


Figure 2. Typical gradients and morphology of Barcelona. The red circle indicates the location of the Raval District.

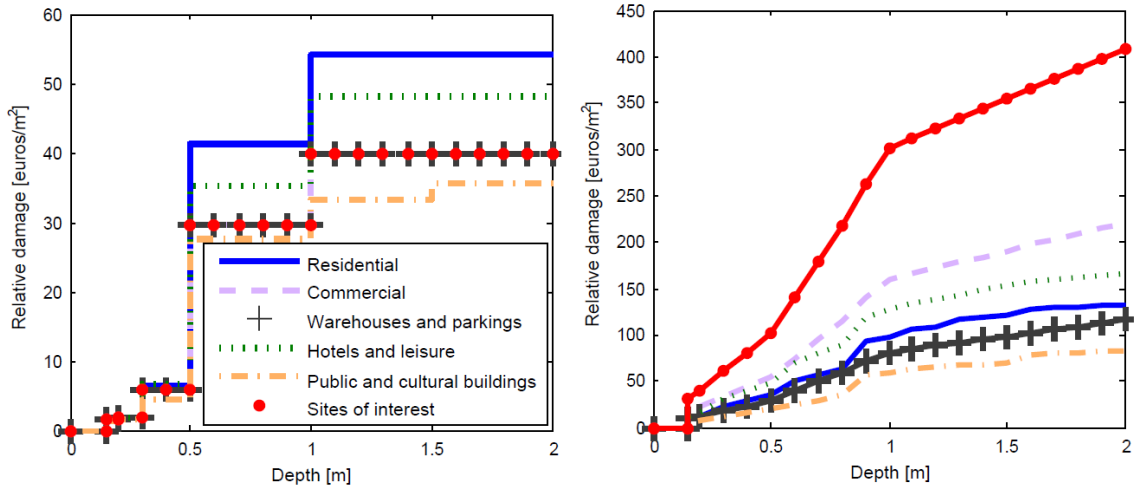


Figure 3. Depth damage curves for the buildings (left) and content (right) taking into account the local conditions of the Raval district.

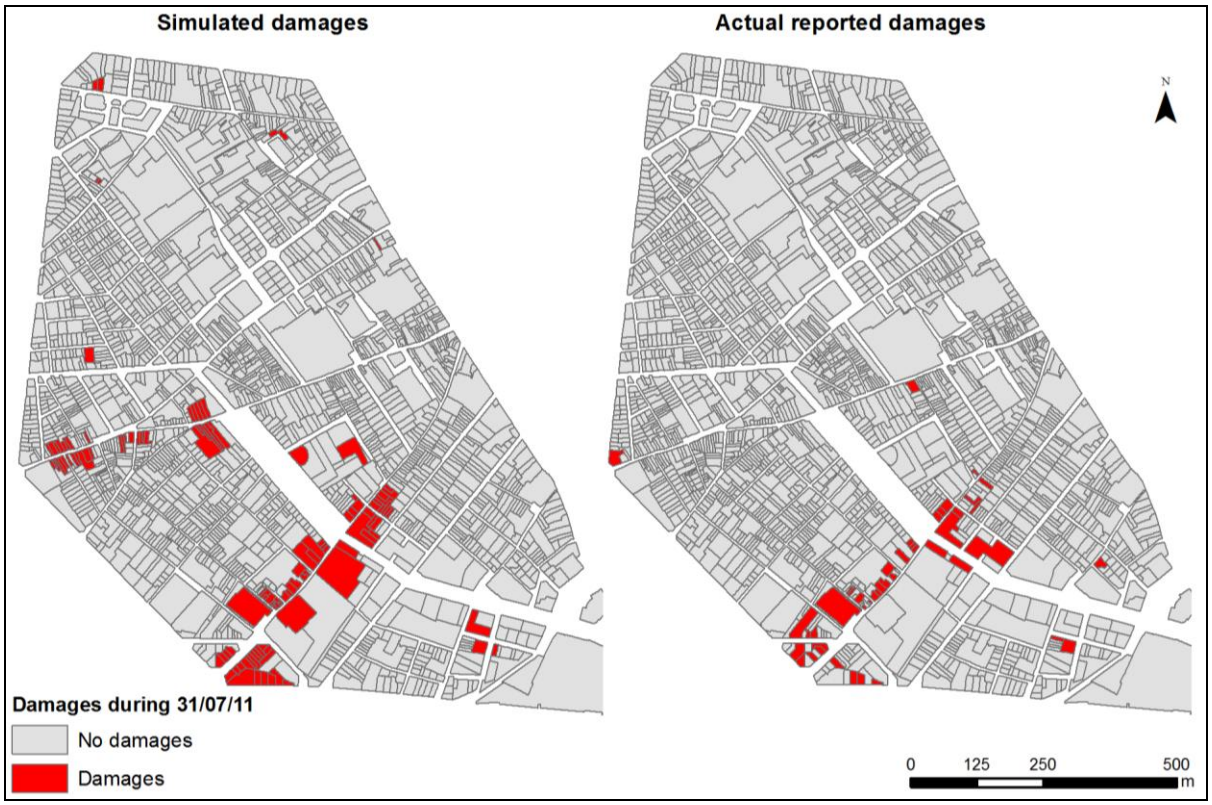


Figure 4. Spatial pattern of damages from the 31/07/2011 flood. Simulated damages (left) present very similar pattern than the actual reported damages of the CCS (right).

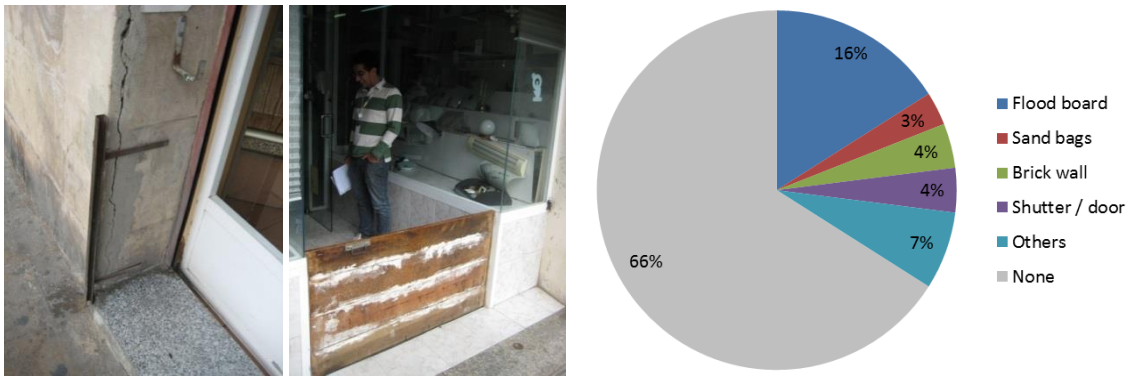


Figure 5. Use of flood protection strategies in the Raval district. On the left and the centre images some details (frame and board) of a sliding wooden panel can be seen. The right graph presents the different types of flood protection strategies used in the Raval district during the 30th July 2011 event.



Figure 6. Land-use classes in the Raval district, the main land-use class of the ground floor is shown at a block scale (left), and vulnerability map, presenting the potential damage of the assets at risk (right).

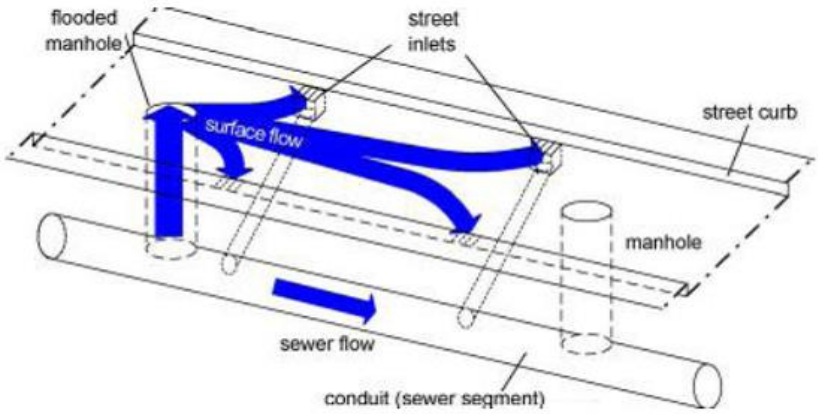


Figure 7. Interaction of surface and sewer flow (dual drainage concept) (Schmitt et al. 2004)



Figure 8. Flood depths in the Raval district for a rain event of return period of 1 year (left), 10 years (centre) and 100 years (right).



Figure 9. Flood damages in the Raval district for a rain event of return period of 1 year (left), 10 years (centre) and 100 years (right).

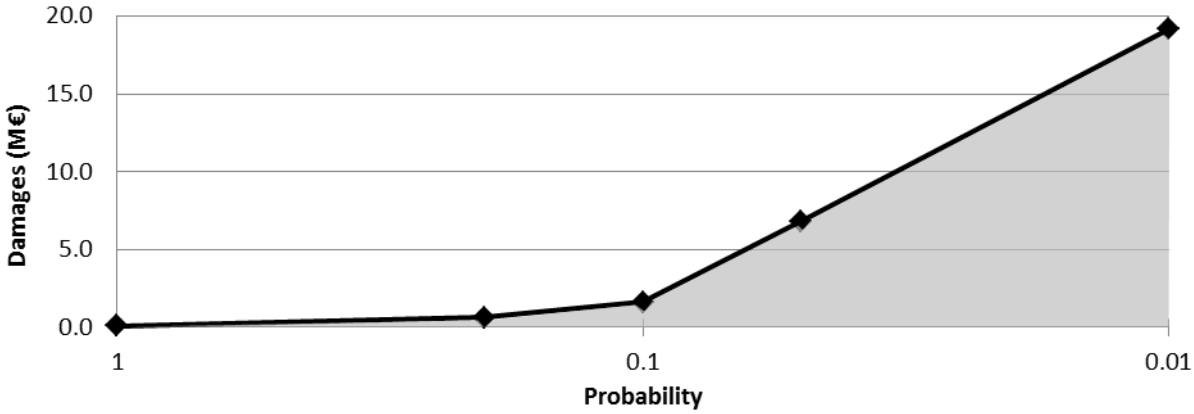


Figure 10. Damage – probability curve for the whole Raval district. The area under this curve expresses the EAD of the region. The probabilities 1, 0.2, 0.1 0.05 and 0.01 represent the events of 1, 5, 10, 20 and 100 years of return period, respectively. The probability axe is expressed in logarithmic scale.